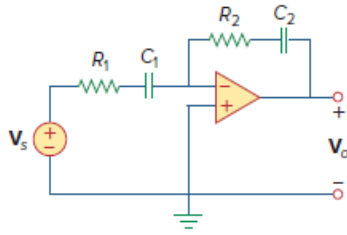


## Problem

Evaluate the voltage gain  $\mathbf{A}_v = \mathbf{V}_o/\mathbf{V}_s$  in the op amp circuit of Fig. 10.117. Find  $\mathbf{A}_v$  at  $\omega = 0$ ,  $\omega \rightarrow \infty$ ,  $\omega = 1/R_1C_1$ , and  $\omega = 1/R_2C_2$ .

Figure 10.117:



## Step-by-step solution

## Step 1 of 7

Refer to Figure 10.117 in the textbook.

Draw the circuit in frequency domain.

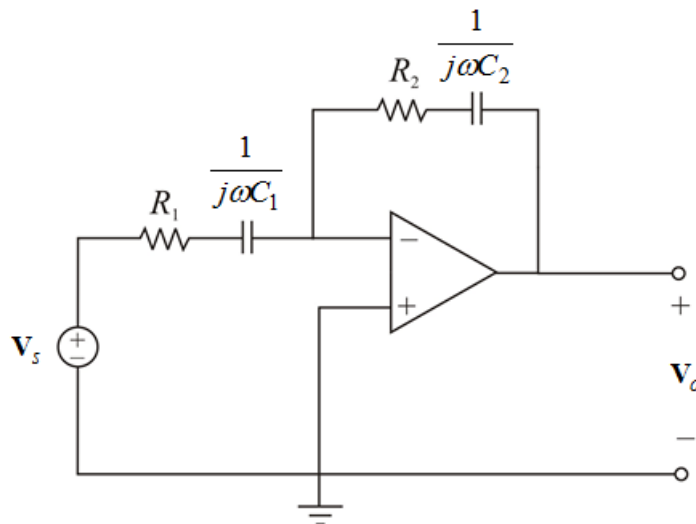


Figure 1

## Step 2 of 7

Calculate the feedback impedance,  $\mathbf{Z}_f$ .

$$\begin{aligned}\mathbf{Z}_f &= R_2 + \frac{1}{j\omega C_2} \\ &= \frac{j\omega R_2 C_2 + 1}{j\omega C_2}\end{aligned}$$

Calculate the input impedance,  $\mathbf{Z}_i$ .

$$\begin{aligned}\mathbf{Z}_i &= R_1 + \frac{1}{j\omega C_1} \\ &= \frac{j\omega R_1 C_1 + 1}{j\omega C_1}\end{aligned}$$

**Step 3** of 7

Calculate the voltage gain of the inverting op-amp.

$$\begin{aligned}
 A_v &= -\frac{Z_f}{Z_i} \\
 &= -\frac{\frac{j\omega R_2 C_2 + 1}{j\omega C_2}}{\frac{j\omega R_1 C_1 + 1}{j\omega C_1}} \\
 &= -\frac{C_1}{C_2} \left( \frac{j\omega R_2 C_2 + 1}{j\omega R_1 C_1 + 1} \right)
 \end{aligned}$$

Therefore, the voltage gain is  $A_v = -\frac{C_1}{C_2} \left( \frac{j\omega R_2 C_2 + 1}{j\omega R_1 C_1 + 1} \right)$

**Step 4** of 7

Calculate the voltage gain at  $\omega = 0$ .

$$A_v = -\frac{C_1}{C_2} \left( \frac{j\omega R_2 C_2 + 1}{j\omega R_1 C_1 + 1} \right)$$

Substitute 0 in  $\omega$ .

$$\begin{aligned}
 A_v &= -\frac{C_1}{C_2} \left( \frac{j(0) R_2 C_2 + 1}{j(0) R_1 C_1 + 1} \right) \\
 &= -\frac{C_1}{C_2} \left( \frac{1}{1} \right) \\
 &= -\frac{C_1}{C_2}
 \end{aligned}$$

Therefore, the voltage gain value at  $\omega = 0$  is  $-\frac{C_1}{C_2}$ .

**Step 5** of 7

Calculate the voltage gain at  $\omega \rightarrow \infty$ .

$$\begin{aligned} \mathbf{A}_v &= -\frac{C_1}{C_2} \left( \frac{j\omega R_2 C_2 + 1}{j\omega R_1 C_1 + 1} \right) \\ &= -\frac{C_1}{C_2} \left( \frac{R_2 C_2 + \frac{1}{j\omega}}{R_1 C_1 + \frac{1}{j\omega}} \right) \end{aligned}$$

Substitute  $\infty$  for  $\omega$ .

$$\begin{aligned} \mathbf{A}_v &= -\frac{C_1}{C_2} \left( \frac{R_2 C_2 + \frac{1}{j(\infty)}}{R_1 C_1 + \frac{1}{j(\infty)}} \right) \\ &= -\frac{C_1}{C_2} \left( \frac{R_2 C_2 + 0}{R_1 C_1 + 0} \right) \\ &= -\frac{R_2}{R_1} \end{aligned}$$

Therefore, the voltage gain value at  $\omega \rightarrow \infty$  is  $\boxed{-\frac{R_2}{R_1}}$ .

#### Step 6 of 7

Calculate the voltage gain at  $\omega = \frac{1}{R_1 C_1}$ .

$$\mathbf{A}_v = -\frac{C_1}{C_2} \left( \frac{j\omega R_2 C_2 + 1}{j\omega R_1 C_1 + 1} \right)$$

Substitute  $\frac{1}{R_1 C_1}$  for  $\omega$ .

$$\begin{aligned} \mathbf{A}_v &= -\frac{C_1}{C_2} \left( \frac{j \left( \frac{1}{R_1 C_1} \right) R_2 C_2 + 1}{j \left( \frac{1}{R_1 C_1} \right) R_1 C_1 + 1} \right) \\ &= -\frac{C_1}{C_2} \left( \frac{1 + j \left( \frac{R_2 C_2}{R_1 C_1} \right)}{1 + j} \right) \end{aligned}$$

Therefore, the voltage gain value at  $\omega = \frac{1}{R_1 C_1}$  is  $\boxed{-\frac{C_1}{C_2} \left( \frac{1 + j \left( \frac{R_2 C_2}{R_1 C_1} \right)}{1 + j} \right)}$ .

#### Step 7 of 7

Calculate the voltage gain at  $\omega = \frac{1}{R_2 C_2}$ .

$$\mathbf{A}_v = -\frac{C_1}{C_2} \left( \frac{j\omega R_2 C_2 + 1}{j\omega R_1 C_1 + 1} \right)$$

Substitute  $\frac{1}{R_2 C_2}$  for  $\omega$ .

$$\begin{aligned} \mathbf{A}_v &= -\frac{C_1}{C_2} \left( \frac{j \left( \frac{1}{R_2 C_2} \right) R_2 C_2 + 1}{j \left( \frac{1}{R_2 C_2} \right) R_1 C_1 + 1} \right) \\ &= -\frac{C_1}{C_2} \left( \frac{1 + j}{1 + j \left( \frac{R_1 C_1}{R_2 C_2} \right)} \right) \end{aligned}$$

Therefore, the voltage gain value at  $\omega = \frac{1}{R_2 C_2}$  is  $\boxed{-\frac{C_1}{C_2} \left( \frac{1 + j}{1 + j \left( \frac{R_1 C_1}{R_2 C_2} \right)} \right)}$ .